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Temporal Concept Dynamics in Diffusion-based Text-to-Audio Generation

Research Project Report

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Chapter 1

Introduction

1.1 Background

Diffusion models have recently become a dominant paradigm for high-quality text-to-audio generation, enabling the synthesis of complex and semantically rich audio from natural language prompts. In music generation, these models can produce nuanced attributes such as tempo, instrumentation, rhythm, and texture with remarkable fidelity. However, despite their empirical success, the internal mechanisms through which such semantic concepts emerge and evolve during the iterative denoising process remain poorly understood.

Diffusion-based generation proceeds through a sequence of gradual refinements in latent space, transforming noise into structured audio representations. While this iterative structure suggests that semantic attributes may form progressively over time, existing work has largely treated the model as a black box, focusing on output quality rather than the dynamics of semantic formation along the denoising trajectory. Consequently, it remains unclear whether different musical concept categories emerge at distinct stages of the diffusion process or whether they are encoded in separable latent patterns.

Recent advances in interpretability for diffusion models attempt to uncover internal semantic structure using approaches such as Concept Bottleneck Models [1, 2] and Sparse Autoencoders [3]. These methods map intermediate latent representations to human-interpretable semantic concepts and typically analyze activations at isolated diffusion timesteps rather than modeling their temporal evolution. Moreover, they often require architectural modifications or additional annotated supervision to train auxiliary components, which may limit faithfulness to the original pretrained model and complicate deployment in generative settings.

In contrast, this project investigates semantic emergence directly within the native diffusion trajectory of a pretrained text-to-audio model, AudioLDM2 [4], without introducing architectural changes (Code in GitHub[1]). By analyzing intermediate latent behavior and introducing controlled prompt-conditioned interventions [5] during sampling, the aim is to better understand how fine-grained

[1] `https://github.com/ShakyaAbeytunge/temporal-concept-dynamics-audioldm2.git`

musical concepts are formed, amplified, and modulated throughout the generative process.

1.2 Research Questions

Despite the growing success of diffusion-based text-to-audio models, it remains unclear how fine-grained musical concepts are represented and shaped during the iterative denoising process. In particular, diffusion models generate audio through a gradual refinement of noisy latent variables, yet the temporal emergence of semantic attributes within this trajectory is not well understood. This project addresses the following core research questions:

1. **Do distinct musical concept categories (e.g., tempo, instrumentation, rhythm) exhibit identifiable patterns in the intermediate diffusion latents during generation?**

2. **At which stages of the denoising process do fine-grained musical concepts become influential, and does this timing differ across concept categories?**

3. **Can controlled modifications of the textual conditioning during sampling reliably steer specific musical attributes, and how does the timing of such modifications affect the resulting generation?**

To answer these questions, I first analyze intermediate latent representations across diffusion timesteps to investigate whether concept-specific divergence patterns emerge between semantically opposing prompts. Due to the limited expressiveness of this latent divergence analysis, a Prompt-Conditioned Intervention (PCI) strategy [5] is employed, which allows dynamic modification of textual conditioning at controlled diffusion timesteps. This intervention-based framework enables us to probe when and how semantic concepts exert influence over the generative trajectory.

By combining latent trajectory analysis with controlled conditioning interventions and quantitative evaluation via CLAP-based [6] similarity metrics, this project aims to provide deeper insight into semantic formation mechanisms in diffusion-based music generation models, while preserving the integrity of the original pretrained architecture.

Chapter 2

Related Work

2.1 Diffusion Models for Text-to-Audio Generation

Diffusion models have emerged as a dominant framework for high-quality generative modeling. Denoising Diffusion Probabilistic Models (DDPMs) [7] introduced the iterative denoising paradigm, in which data are generated by progressively refining Gaussian noise through a learned reverse process. Subsequent advances, such as Denoising Diffusion Implicit Models (DDIM) [8] improved sampling efficiency, while latent diffusion models [9] demonstrated that performing diffusion in a compressed latent space significantly reduces computational cost while preserving generation quality.

In the audio domain, diffusion-based approaches have enabled semantically aligned text-to-audio synthesis by conditioning the denoising process on textual embeddings [10, 11, 4]. Among these systems, AudioLDM2 [4] provides a unified framework for text-conditioned audio and music generation by operating in a latent space derived from pretrained audio representations and leveraging cross-modal alignment mechanisms. Its latent diffusion formulation makes it computationally efficient while maintaining strong semantic controllability.

2.2 Interpretability in Diffusion Models

Recent work has begun investigating interpretability in diffusion models using approaches such as Concept Bottleneck Models [1] and Sparse Autoencoders [12], which aim to map latent activations to human-interpretable semantic concepts. While these methods reveal meaningful structure within intermediate representations, they typically analyze isolated timesteps rather than modeling semantic evolution across the full denoising trajectory. Moreover, these approaches introduce architectural modifications or auxiliary supervision, potentially affecting the original model dynamics. Consequently, a gap remains in understanding how semantic concepts develop temporally within the native diffusion process.

2.3 Concept Control in Diffusion

A growing line of research investigates how semantic concepts can be controlled within diffusion-based generative models. In many generative systems, textual prompts act as high-level conditioning signals that guide the overall generation process. However, the internal mechanisms through which these semantic concepts influence the denoising trajectory remain only partially understood [5]. Understanding how different attributes emerge and interact during generation is therefore an important step toward improving controllability in diffusion models.

Recent work in the image generation domain has shown that semantic concepts may emerge at different stages of the diffusion process, suggesting that the timing of conditioning can play a critical role in shaping the generated output. For example, Prompt-to-Prompt methods demonstrate that modifying attention maps during diffusion can selectively alter specific visual attributes while preserving the rest of the image [13]. More recently, Gorgun et al. [5] introduced the Prompt-Conditioned Intervention (PCI) framework, which studies how concept-specific interventions at different diffusion timesteps affect the presence of semantic attributes in generated images. Their findings suggest that different concepts may have distinct temporal activation patterns within the diffusion process.

Although these insights highlight the importance of temporal dynamics in diffusion models, most existing studies focus on the image domain. In the context of music generation, where audio signals exhibit complex temporal structures involving rhythm, timbre, and harmonic content, the emergence and interaction of semantic concepts remain largely unexplored. Investigating how musical attributes evolve during the diffusion process could therefore provide valuable insights for improving controllability and enabling more reliable editing mechanisms in diffusion-based music generation models.

Chapter 3

Methodology

This section consists of two main parts. In Section 3.1, the analysis of intermediate diffusion latent patterns for different musical concepts is explained. Due to the limited expressive power of this analysis alone, I extend my approach in Section 3.2 by adapting the Prompt-Conditioned Interventions (PCI) framework, originally introduced for text-to-image generation [5], to the text-to-audio setting.

The following terminology will be used throughout the remainder of this report.

3.0.1 Concept

A *concept* refers to a semantically meaningful musical attribute that can be expressed through natural language and potentially manifested in generated audio. Concepts may describe properties such as instrumentation (e.g., violin), tempo (e.g., fast), acoustic environment (e.g., concert hall), or performance characteristics (e.g., live recording). In this work, a concept is operationally defined as a textual token or phrase whose presence in the prompt is expected to induce a measurable change in the generated audio.

3.0.2 Concept Category

A *concept category* is a higher-level grouping of semantically related concepts that describe a common musical dimension. For example, instrumentation, tempo, acoustic environment, and performance style constitute different concept categories. Each category defines a controlled semantic axis along which variations are studied. Structuring concepts into categories allows systematic evaluation by restricting comparisons to competing concepts within the same semantic dimension.

3.0.3 Fine-grained Concepts

Fine-grained concepts are the specific, concrete instances within a concept category. They represent mutually comparable alternatives along the same semantic axis. For example, within the instrumentation category, fine-grained concepts include instruments such as trumpet, violin, piano, and flute. In the tempo category, examples may include slow, moderate, or fast. Fine-grained concepts are used for controlled prompt conditioning and quantitative evaluation, ensuring that similarity comparisons are performed only among semantically related alternatives.

3.1 Latent Trajectory Analysis in Diffusion

The objective of this analysis is to investigate whether concept-specific patterns emerge in the intermediate latent representations of the diffusion process for different musical concept categories (e.g., tempo, instrumentation, rhythm, acoustic environment).

For each concept category, I constructed sets of prompt pairs in which the two prompts differed only by opposing fine-grained concepts within the same category. For example, in the tempo category, a pair can be:

- Prompt A: "A *fast* ambient music"
- Prompt B: "A *slow* ambient music"

Both prompts were used to generate audio samples while recording the intermediate latent representations at each denoising timestep. Let $\mathbf{z}_t^A$ and $\mathbf{z}_t^B$ denote the latent variables at timestep t for Prompt A and Prompt B, respectively. To quantify divergence between the two diffusion trajectories, I computed the ℓ_2 distance at each timestep:

$$D_t = \|\mathbf{z}_t^A - \mathbf{z}_t^B\|_2. \tag{3.1}$$

This procedure produced divergence curves $\{D_t\}_{t=1}^{T}$ across the denoising process. The resulting curves were analyzed for multiple prompt pairs within the same concept category, as well as across different concept categories, in order to identify consistent structural patterns in latent space evolution.

However, as discussed in Section 4.1, this analysis did not reveal stable, concept-specific signatures that could reliably distinguish between concepts. The latent divergence patterns were insufficiently discriminative across concept categories. Consequently, I moved toward a more expressive and intervention-based approach, described in the following section.

3.2 Prompt Condition Intervention (PCI)

Prompt Conditioned Intervention (PCI) is a method adapted from recent advances in text-to-image generation [5], which aims to systematically control and evaluate the influence of specific concepts on the generative process. In this context, PCI is used to intervene in music generation by explicitly inserting fine-grained concepts, such as specific instruments, rhythmic patterns, or musical styles, into a base prompt. This framework allows us to investigate not only whether the generative model can faithfully incorporate these concepts, but also when during the generation process they exert the strongest influence.

The PCI framework is built upon carefully designed *base prompts* that establish a controlled musical context. By appending structured concept-specific phrases to these base prompts, prompt pairs are constructed that differ only in the inclusion of a single target concept, referred to as *concept-conditioned prompts*.

An illustrative example for the concept category *genre* is given below:

- **Base Prompt:** "A music track"
- **Concept-Conditioned Prompt:** "A *jazz* music track"

PCI operates by intervening in the diffusion process at a designated time step τ. Specifically, generation is first initialized using the base prompt, allowing the model to establish a coherent global musical structure. At a chosen diffusion step τ, the conditioning is replaced with the concept-conditioned prompt, thereby introducing the target concept into an already partially formed representation. From this point onward, the denoising process proceeds under the modified conditioning. By varying τ across the diffusion trajectory, it can be systematically analyzed how the timing of the intervention affects the model's ability to incorporate the target concept. This temporal manipulation enables us to study not only concept controllability, but also the sensitivity of the generative process to late-stage versus early-stage conditioning.

3.2.1 Concept Categories

I initially considered four overarching concept categories comprising multiple fine-grained concepts:

- **Instrumentation:** violin, trumpet, piano, guitar, flute, drums, etc.
- **Genre:** jazz, classical, electronic dance music, pop, hip hop, etc.
- **Tempo:** fast, moderate, slow
- **Mood:** happy, sad, energetic, calm

For each fine-grained concept within these categories, I generated audio samples using AudioLDM2 ("audioldm_48k" model) to assess baseline controllability. However, I observed that the model struggled to reliably generate mood-related concepts—particularly *sad* music—as well as tempo-related concepts. In many cases, prompts specifying *fast* or *slow* music did not produce clearly distinguishable temporal characteristics.

Moreover, both mood and tempo are inherently subjective attributes, making their evaluation more ambiguous. This ambiguity was also reflected in the CLAP audio–text similarity scores, which did not provide consistently convincing alignment for these categories in the initial experiments.

Based on these observations, the subsequent PCI analysis was restricted to two more structurally grounded categories: **instrumentation** and **genre**.

3.2.2 Base Prompts

To ensure controlled experimental conditions, I first conducted a series of preliminary generations using multiple candidate base prompts combined with fine-grained concepts from different concept categories. The goal of this qualitative analysis was to examine how strongly the base prompt context itself influenced the presence of specific musical attributes.

For each concept category, several generic base prompts were evaluated to determine whether they implicitly biased the generation toward certain concepts. This preliminary investigation revealed that some prompts inherently suggested particular musical characteristics, thereby reducing the ability to isolate the effect of explicit concept injection. Consequently, base prompts were selected carefully to provide distinguishable yet controlled musical contexts while minimizing unintended concept bias.

Instrumentation. For the instrumentation category, candidate base prompts included: "An instrumental music piece," "A music track played by an orchestra," "A solo musical performance," and "A live recording of a band." The fine-grained concepts consisted of individual instruments such as trumpet, violin, piano, drums, flute, guitar, harp, and saxophone. Qualitative inspection indicated that certain base prompts strongly implied specific instruments—for example, orchestral settings often suggested string instruments such as violin—thereby diminishing the measurable impact of explicit instrument insertion.

To balance contextual richness with flexibility for concept intervention, two base prompts were ultimately selected: "A music track played by an orchestra" and "A live recording of a band." These prompts provide structurally distinct musical environments while remaining sufficiently open to controlled concept injection.

Importantly, for each base prompt I explored instrument concepts that are not trivially implied by the context. For example, within the orchestral setting, I investigated instruments such as piano and drums, which are not always dominant or explicitly featured in traditional orchestral arrangements. Conversely, for the band setting, I examined instruments such as violin, trumpet, and flute, which are less prototypical in standard band configurations. This deliberate cross-context selection reduces prior prompt bias and enables a more precise evaluation of concept controllability.

Genre. In contrast, for the genre category, the musical identity is primarily determined by the genre descriptor itself. To avoid contextual bias, a minimal base prompt—"A music track"—was adopted. The corresponding concept-conditioned template was defined as "A {genre} music track," ensuring that genre identity arises explicitly from the injected concept rather than from prior contextual cues.

However, preliminary generations using the base prompt "A music track" revealed a noticeable bias in the model's default outputs. Quantitative analysis using CLAP embeddings showed that many base generations exhibited relatively high similarity to pop and rock genres. This inherent bias makes it difficult to isolate the effect of explicit genre injection for these categories, as they already tend to emerge in the base prompt generations.

To mitigate this issue, I selected a set of less dominant and more distinguishable genres for analysis: jazz, classical, hip hop, and electronic dance music (EDM). These genres exhibit clearer boundaries in the embedding space, allowing for more reliable measurement of concept activation under PCI.

3.2.3 Seed Selection

After selecting appropriate base prompts for each concept category, concept-conditioned prompts were constructed by appending a structured phrase containing a fine-grained concept (e.g., "with a $<$ *concept* $>$ solo" in the case of instrumentation). Let p_b denote a selected base prompt and $c \in \mathcal{C}$ a fine-grained concept from the concept set of a given category. For each seed $s \in \{0, 1, \ldots, 200\}$, a base audio sample was generated as

$$x_b^{(s)} = G(p_b, s), \tag{3.2}$$

where $G(\cdot, s)$ denotes the text-to-audio generation process with fixed parameters and deterministic seed s. For each concept c, a concept-conditioned sample was generated using the same seed:

$$x_{b,c}^{(s)} = G(p_b \oplus c, s), \tag{3.3}$$

where $p_b \oplus c$ represents the base prompt augmented with the concept phrase. Keeping all generation parameters identical ensures that any differences between $x_b^{(s)}$ and $x_{b,c}^{(s)}$ are attributable solely to prompt conditioning rather than stochastic variation.

To quantitatively evaluate concept presence, CLAP audio embeddings were extracted for each generated waveform. Let $f_a(\cdot)$ denote the CLAP audio encoder and $f_t(\cdot)$ the CLAP text encoder. The audio embedding of a generated sample and the text embedding of a concept are given by

$$\mathbf{a}_{b,c}^{(s)} = f_a(x_{b,c}^{(s)}), \qquad \mathbf{t}_k = f_t(k), \quad k \in \mathcal{C}. \tag{3.4}$$

Concept alignment was measured using cosine similarity between the CLAP audio embedding $\mathbf{a}$ of a generated sample and the text embedding $\mathbf{t}_k$ of concept k:

$$\mathrm{Sim}(x, k) = \frac{\mathbf{a} \cdot \mathbf{t}_k}{\|\mathbf{a}\| \, \|\mathbf{t}_k\|}. \tag{3.5}$$

To isolate the effect of concept injection, a concept activation score was defined as the similarity difference between the concept-conditioned generation and the corresponding base generation:

$$\Delta_k^{(s)} = \mathrm{Sim}(x_{b,c}^{(s)}, k) - \mathrm{Sim}(x_b^{(s)}, k), \tag{3.6}$$

where $x_b^{(s)}$ denotes the generation obtained from the base prompt using seed s, and $x_{b,c}^{(s)}$ denotes the generation obtained after injecting concept c.

A seed s was considered a *potentially good candidate* for concept c if it satisfied

$$\arg\max_{k \in \mathcal{C}} \Delta_k^{(s)} = c \tag{3.7}$$

and

$$\Delta_c^{(s)} > \tau_1, \tag{3.8}$$

where τ_1 is an experimentally determined positive threshold specific to each concept subcategory (e.g., $\tau_1 = 0.05$ for instrumentation). This condition ensures that the injected concept produces both the strongest relative similarity increase among competing concepts and a sufficiently large absolute improvement over the base generation.

To further ensure that the concept is not already strongly present in the base prompt generation, an additional filtering step was introduced. For each seed s satisfying the above criteria, I computed the CLAP similarity between the base generation and the concept text:

$$\mathrm{Sim}(x_b^{(s)}, c). \tag{3.9}$$

Seeds were retained only if

$$\mathrm{Sim}(x_b^{(s)}, c) < \tau_2, \tag{3.10}$$

where τ_2 is another experimentally chosen threshold (set to $\tau_2 = 0.05$ in experiments). This constraint ensures that the concept is minimally aligned with the base generation prior to intervention, thereby reducing confounding effects.

Let $\mathcal{S}_{\text{filtered}}$ denote the set of seeds satisfying both the activation criteria (τ_1) and the base-similarity constraint (τ_2). The final seeds were selected as the $top - N$ seeds with the lowest base-prompt similarity to the concept text:

$$\mathcal{S}_{\text{final}} = \arg\min_{s \in \mathcal{S}_{\text{filtered}}} \mathrm{Sim}(x_b^{(s)}, c). \tag{3.11}$$

This two-stage filtering procedure ensures that selected seeds (i) exhibit strong and concept-specific activation after intervention, and (ii) have minimal prior alignment with the injected concept, allowing clearer attribution of observed effects to the PCI.

To improve computational efficiency, an early stopping strategy was applied during seed evaluation. Once a sufficient number of seeds satisfying both the activation criterion ($\Delta_c^{(s)} > \tau_1$ with correct concept dominance) and the base-similarity constraint ($\mathrm{Sim}(x_b^{(s)}, c) < \tau_2$) was collected for a given concept, further evaluation for that concept was terminated. This prevented unnecessary computation after an adequate candidate pool had been formed.

3.2.4 Evaluation Metrics

After selecting the final seed set $\mathcal{S}_{\text{final}}$ for each concept c, I evaluated the temporal behavior of Prompt Conditioned Intervention (PCI). Unlike the full concept-conditioned generation $x_{b,c}^{(s)} = G(p_b \oplus c, s)$, PCI introduces the concept at a designated diffusion step τ. Let

$$x_{b,c,\tau}^{(s)} \tag{3.12}$$

denote the audio generated using seed s, where the generation is initialized with base prompt p_b and the concept c is injected at diffusion step τ.

Concept Similarity. For each intervention step τ, concept alignment is measured using cosine similarity:

$$\mathrm{Sim}(x_{b,c,\tau}^{(s)}, c) = \frac{f_a(x_{b,c,\tau}^{(s)}) \cdot f_t(c)}{\|f_a(x_{b,c,\tau}^{(s)})\| \, \|f_t(c)\|}. \tag{3.13}$$

This metric quantifies the absolute alignment between the PCI-generated audio and the injected concept.

Activation Score. To isolate the effect of temporal intervention relative to the base generation, a τ-dependent activation score is defined:

$$\Delta_{c,\tau}^{(s)} = \mathrm{Sim}(x_{b,c,\tau}^{(s)}, c) - \mathrm{Sim}(x_b^{(s)}, c). \tag{3.14}$$

Positive values indicate that the intervention at step τ increases concept alignment compared to the base generation.

Mean Activation Curve. For each concept c, similarity and activation scores are averaged across the selected seed set:

$$\overline{\Delta}_c(\tau) = \frac{1}{|\mathcal{S}_{\text{final}}|} \sum_{s \in \mathcal{S}_{\text{final}}} \Delta_{c,\tau}^{(s)}. \tag{3.15}$$

This produces a temporal activation curve that characterizes how the timing of intervention influences concept incorporation.

Success Rate. In addition to continuous activation scores, a binary success indicator is defined:

$$\mathbb{I}_{c,\tau}^{(s)} = \begin{cases} 1 & \text{if } \Delta_{c,\tau}^{(s)} > \delta_c, \\ 0 & \text{otherwise,} \end{cases} \tag{3.16}$$

where δ_c is a concept-specific activation threshold.

To determine δ_c, first, the reference similarity distribution is computed from full concept-conditioned generations without temporal intervention. Let

$$\mathcal{A}_c = \left\{ \mathrm{Sim}\left(x_{b,c}^{(s)}, c\right) \;\middle|\; s \in \mathcal{S}_{\text{final}} \right\} \tag{3.17}$$

denote the set of cosine similarities between the concept text and the fully conditioned generations $x_{b,c}^{(s)} = G(p_b \oplus c, s)$.

The threshold is defined as a fraction of the median reference similarity:

$$\delta_c = \alpha \cdot \mathrm{median}(\mathcal{A}_c), \quad \alpha = 0.25. \tag{3.18}$$

This normalization ensures that the success criterion adapts to the intrinsic detectability of each concept while remaining comparable across intervention steps.

The success rate at step τ is then computed as

$$R_c(\tau) = \frac{1}{|\mathcal{S}_{\text{final}}|} \sum_{s \in \mathcal{S}_{\text{final}}} \mathbb{I}_{c,\tau}^{(s)}. \tag{3.19}$$

This metric captures the proportion of generations in which PCI produces a meaningful improvement in concept alignment relative to the base prompt.

Temporal Sensitivity Analysis. By varying τ across the diffusion trajectory, I obtain similarity, activation, and success curves as functions of intervention time. This enables analysis of when the generative process is most sensitive to concept injection, thereby revealing the temporal dynamics of controllability in diffusion-based music generation.

Base Similarity. For each intervention step τ, the similarity between the base prompt generation (without intervention) and the PCI-conditioned generation is measured:

$$\mathrm{Sim}_{\text{base}}(x_{b,c,\tau}^{(s)}, x_b^{(s)}) = \frac{f_a(x_{b,c,\tau}^{(s)}) \cdot f_a(x_b^{(s)})}{\|f_a(x_{b,c,\tau}^{(s)})\| \, \|f_a(x_b^{(s)})\|}. \tag{3.20}$$

This metric quantifies how closely the intervened generation resembles the original base generation.

Mean Base Similarity Curve. For each concept c, the base similarity is averaged across the selected seed set:

$$\overline{\mathrm{Sim}}_{\mathrm{base},c}(\tau) = \frac{1}{|\mathcal{S}_{\mathrm{final}}|} \sum_{s \in \mathcal{S}_{\mathrm{final}}} \mathrm{Sim}_{\mathrm{base}}(x_{b,c,\tau}^{(s)}, x_b^{(s)}). \tag{3.21}$$

This produces a temporal base-similarity curve that characterizes how strongly the PCI-conditioned generation preserves the original audio content across intervention times.

Temporal Editability Analysis. For ease of interpretation, the mean base similarity curves highlight diffusion time-step regions in green and red to indicate highly successful ($R_c(\tau) > 0.8$) and highly unsuccessful ($R_c(\tau) < 0.2$) intervention regions, respectively. This visualization allows us to assess, for each τ, the extent to which the PCI-conditioned generation deviates from the base generation while still successfully incorporating the injected concept. Consequently, it can be analyzed to determine the effectiveness of temporal intervention in editing the audio to introduce new concepts while preserving the underlying structure of the original generation.

Chapter 4

Experiments & Results

In this project, I adopt AudioLDM2 [4] as the base model for the experiments for three primary reasons. First, its latent diffusion architecture provides direct access to intermediate denoising states, enabling analysis of semantic evolution across timesteps. Second, the model supports flexible text conditioning without requiring architectural modification, making it suitable for controlled intervention experiments. Third, as a publicly available pretrained system, it allows investigation of semantic dynamics without retraining, preserving the integrity of the original model behavior.

Sections 4.1 and 4.2 present the experiments and results of the two approaches employed to analyze the temporal concept dynamics of music generation in AudioLDM2.

4.1 Latent Trajectory Analysis

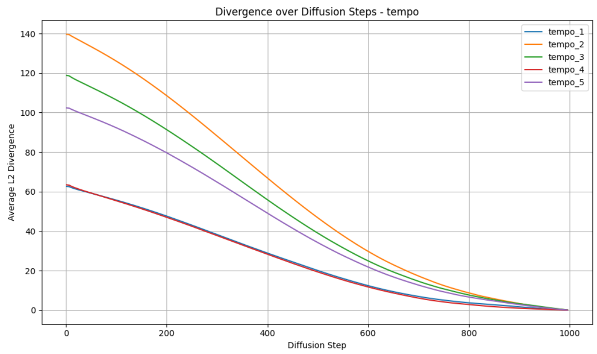
Three concept categories were analyzed: tempo, instrumentation, and rhythm. For each category, five prompt pairs were constructed such that each pair was identical except for a contrastive modification within the targeted concept category. The complete list of prompt pairs is provided in Appendix .1.

For all prompt pairs, the random state of the generation was kept identical to ensure fair comparison. Figure 4.1 illustrates the divergence curves for the prompt pairs. Across all concept categories, the curves exhibit smooth progression toward their final generations and share similar overall shapes, differing primarily in divergence magnitude.

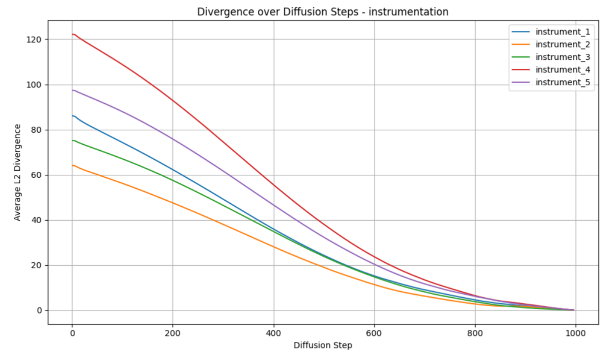
Figure 4.2 presents the mean divergence curves computed from the five prompt pairs within each concept category. While rhythmic concepts exhibit higher variability, reflected by the larger plotted standard deviations, the instrumentation concepts show the lowest variability.

Subfigure 4.2d overlays the mean divergence curves of all concept categories in a single plot. The similarity in curve shapes suggests that intermediate latent divergence does not exhibit concept-specific dynamics during the generation process.

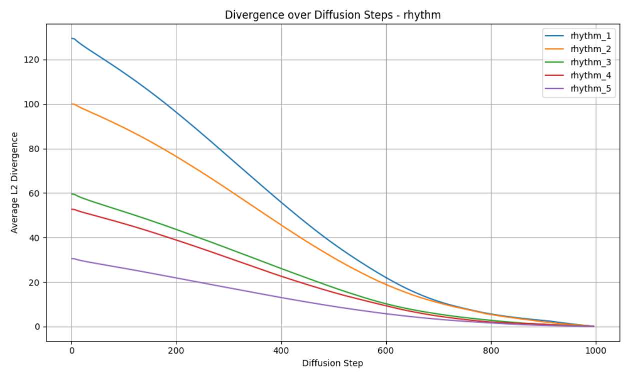
To quantify this visual similarity, I computed pairwise correlations between the divergence curves of prompt pairs within each concept category, as well as across different concept categories. The mean



(a) Tempo concepts



(b) Instrumentation concepts



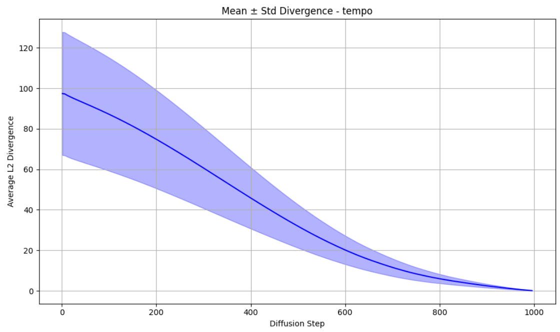
(c) Rhythmic concepts

Figure 4.1: Divergence curves (eq.3.1) across diffusion steps for prompt pairs by concept category.
Note: Diffusion Step 0 indicates the final generation step, and 1000 indicates the start of the generation.

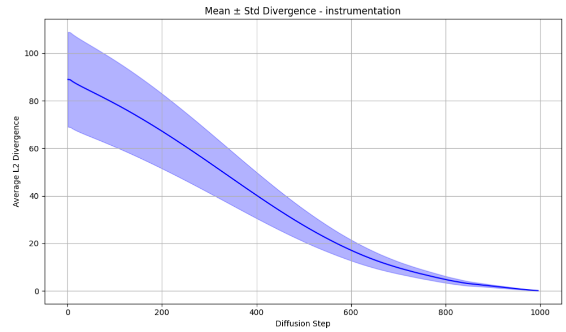
correlation between divergence curves was nearly perfect for all categories: tempo (1.0), instrumentation (0.99), and rhythm (1.0). The within-concept and between-concept average correlations were both 1.0.

These results indicate that there is no significant difference in diffusion behavior across concept categories, nor any specific diffusion time step that responds differently depending on the concept type. This can be attributed to the fact that concept-conditioned prompts are provided from the beginning of the generation process, allowing the model to progressively realize the target concept throughout the diffusion trajectory. In other words, the model is aware of the final objective from the first step and refines toward it incrementally.

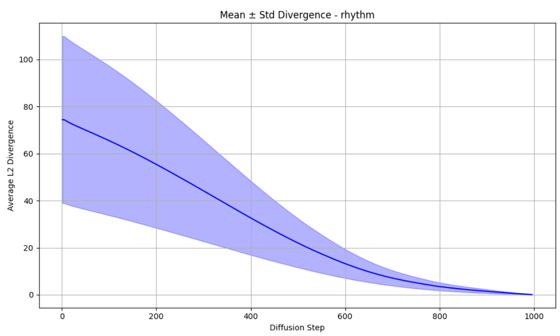
To meaningfully analyze the nuanced temporal dynamics of different concepts, the model should not be conditioned on the final objective at the beginning. Prompt Conditioned Intervention (PCI) addresses this by introducing the concept during intermediate diffusion steps, thereby enabling controlled temporal analysis.



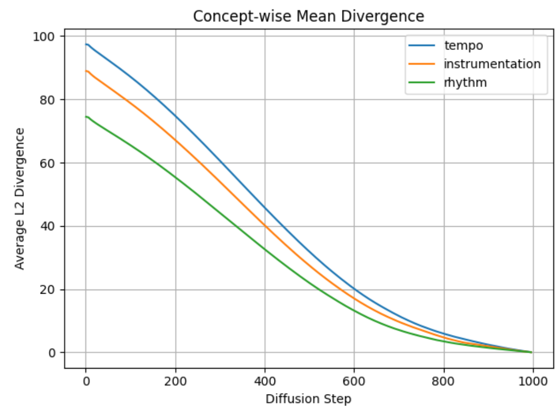
(a) Tempo concepts



(b) Instrumentation concepts



(c) Rhythmic concepts



(d) All concept categories

Figure 4.2: Mean divergence curves across diffusion steps for concept categories.
Note: Diffusion Step 0 indicates the final generation step, and 1000 indicates the start of the generation.

4.2 Prompt Condition Intervention (PCI)

4.2.1 Experimental Setup

To systematically evaluate the effectiveness of Prompt Conditioned Intervention (PCI), I conducted controlled generation experiments using the AudioLDM2 model ("audioldm_48k"), which employs a diffusion process with 1000 denoising steps. Given computational constraints and the experimental time frame, I focused on the top 10 seeds per concept, as determined by the seed selection procedure described in Section 3.2.3.

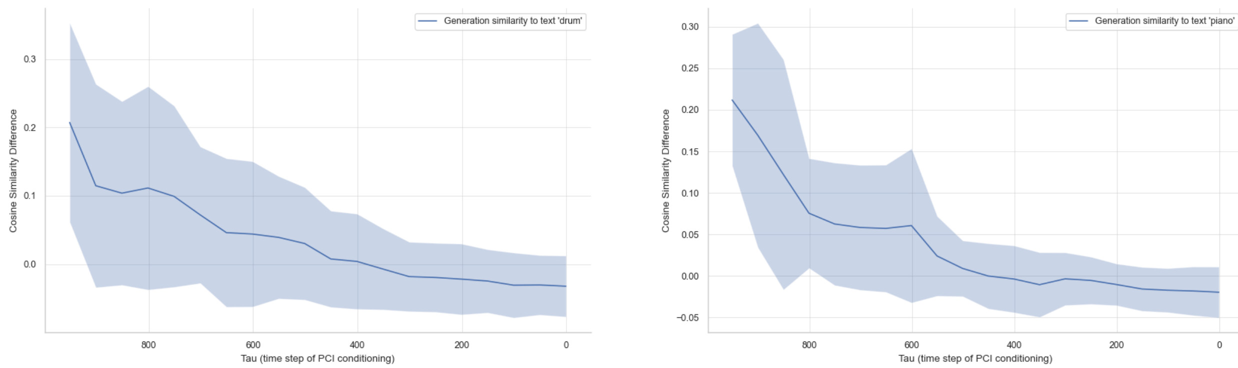
For each concept, the PCI intervention was applied at 50 evenly spaced diffusion steps, corresponding to 20 (τ) increments spanning the full range of 0 to 1000. Two concept categories—instrumentation and genre—were explored. Overall, this setup resulted in a total of 1800 generated audio samples, calculated as 9 fine-grained concepts (5 instruments and 4 genres) $\times$ 20 tau steps $\times$ 10 seeds $= 1800$. Each sample was analyzed using CLAP embeddings to quantify the alignment of the generated audio with the target concept.

This configuration allows us to track concept activation dynamics across both seeds and time steps, providing insight into when and how PCI effectively amplifies specific musical attributes in diffusion-based audio generation.

4.2.2 Results

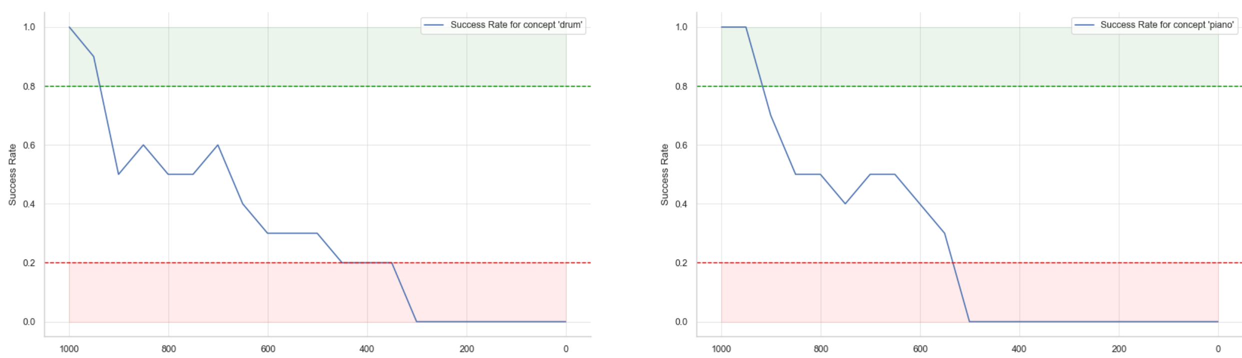
The results of PCI were evaluated from two complementary perspectives across different intervention time steps (τ): first, the **sensitivity** of concept injection; and second, the extent to which PCI facilitates **editability** of the generated audio.

Temporal Sensitivity Analysis



(a) Caption 'drum' (b) Caption 'piano'

Figure 4.3: Mean activation curves (eq.3.14 and eq.3.15) across diffusion steps for instrument concepts in an *orchestral* context.



(a) Caption 'drum' (b) Caption 'piano'

Figure 4.4: Success Rate curves across diffusion steps for instrument concepts in an *orchestral* context.

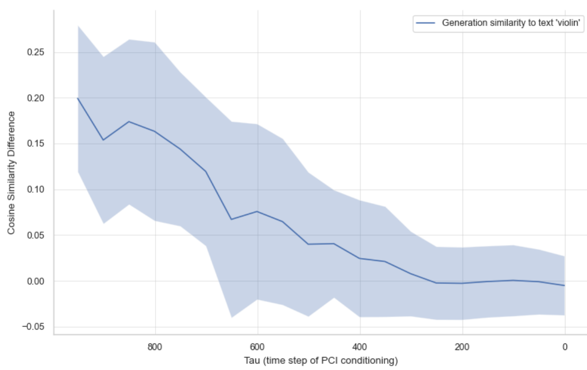
Instrumentation. For all considered instruments, the mean activation curves (Figures 4.3 and 4.5) exhibit an overall decreasing trend as a function of the intervention step τ, with only minor local fluctuations at a few time steps. This behavior indicates that interventions applied at later diffusion steps (i.e., as reversed, for smaller τ) yield diminishing improvements in concept alignment relative to the base generation $x_b^{(s)}$. In other words, the effectiveness of Prompt Conditioned Intervention in introducing the target concept c decreases as the intervention is delayed in the generation process.

Success rate curves (Figures 4.4 and 4.6), where the success rate is obtained by thresholding the activation score, provide a clearer view of concept-specific behavior during the intervention process.

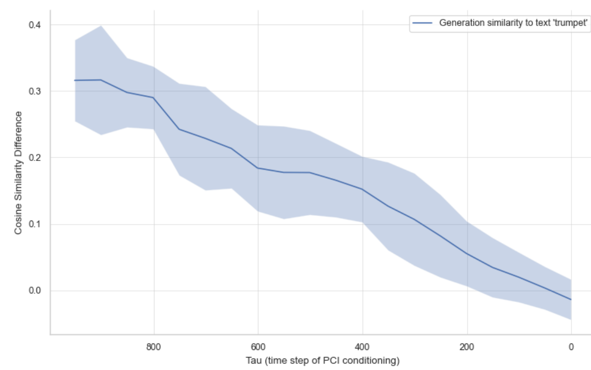
I highlight regions in the plots where $R_c(\tau) > 0.8$ (green) as highly successful and $R_c(\tau) < 0.2$ (red) as highly unsuccessful.

It is observed that for all concepts except *trumpet*, the intervention is largely unsuccessful for $\tau < 400 \pm 50$. In contrast, the *trumpet* concept maintains high success rates over a wider range of τ, specifically up to $\tau \approx 350$. For *piano* and *drum*, the effective intervention window is relatively narrow, with the success rate dropping towards $R_c(\tau) \approx 0.5$ for $\tau < 950$.

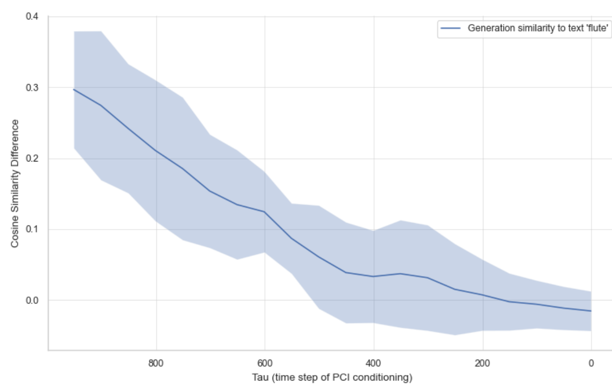
Considering the averaged curves across all instrument concepts (Figures 4.7 and 4.8), the *band* context shows improved concept insertability up to $\tau \approx 700$, whereas the *orchestral* context remains effective up to $\tau \approx 950$. This improvement in the band context is mainly driven by the strong insertability of the *trumpet* concept.



(a) Concept 'violin'



(b) Concept 'trumpet'

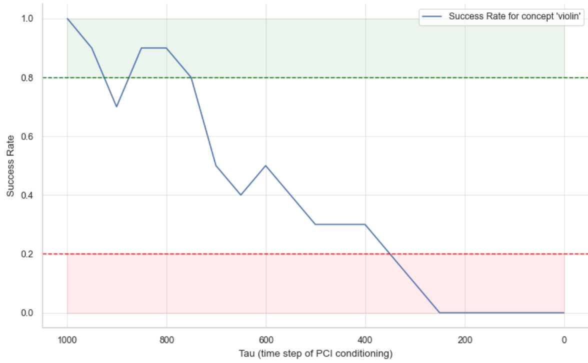


(c) Concept 'flute'

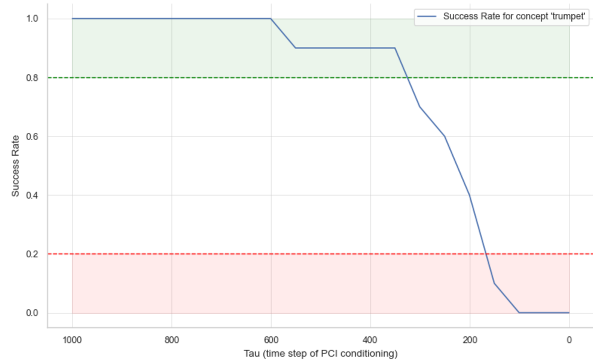
Figure 4.5: Mean activation curves (eq.3.14 and eq.3.15) across diffusion steps for instrument concepts in a *band* context.

Genre. The *classical* genre concept shows a smooth decrease in activation scores as τ decreases (Figure 4.9) compared to the other genre concepts. The *jazz* concept exhibits a relatively flat curve until $\tau \approx 500$, while the *EDM* and *hip hop* curves are largely similar, showing an overall declining trend with a few local fluctuations. Overall for genre concepts, later interventions lead to reduced concept visibility (Figure 4.11 left).

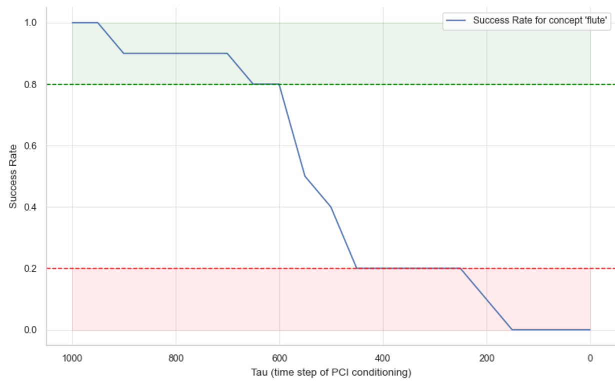
Considering the success rates (Figure 4.10, the *EDM* and *jazz* concepts exhibit a relatively narrow



(a) Concept 'violin'

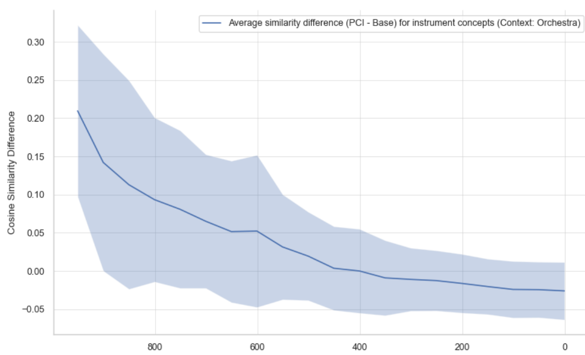


(b) Concept 'trumpet'

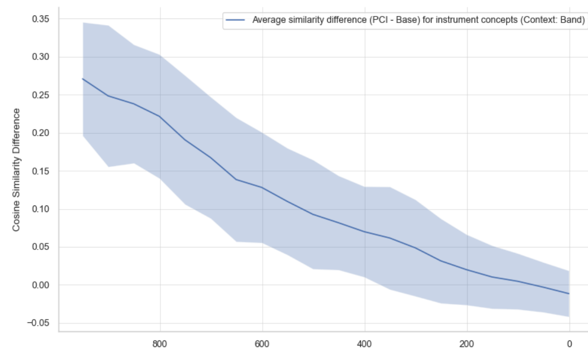


(c) Concept 'flute'

Figure 4.6: Success Rate curves across diffusion steps for instrument concepts in a *band* context.



(a) Orchestral context

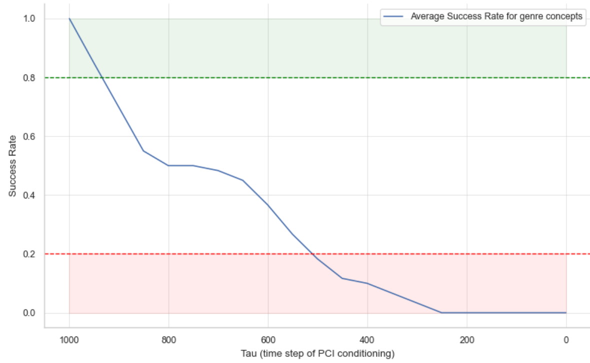


(b) Band context

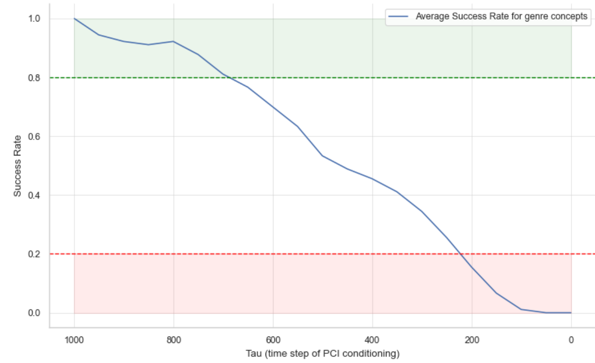
Figure 4.7: Mean activation curves across diffusion steps, averaged across instrument concepts

window of high success, primarily for $\tau > 950$. Beyond this region, *jazz* maintains a comparatively higher success rate, whereas the success rate of *EDM* declines toward $R_c(\tau) \approx 0.5$. In terms of failure regions, *EDM* reaches low success rates earlier than all other concepts, while *jazz* becomes highly unsuccessful only at much later intervention stages (for $\tau < 200$).

By contrast, the *classical* and *hip hop* concepts exhibit a substantially wider high-success window, extending up to $\tau \approx 600$, and become highly unsuccessful for $\tau < 250$. Overall, for the genre concepts

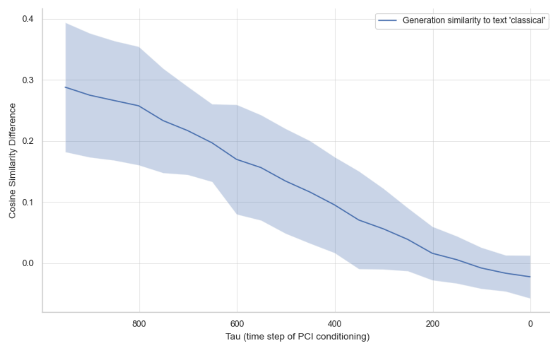


(a) Orchestral context

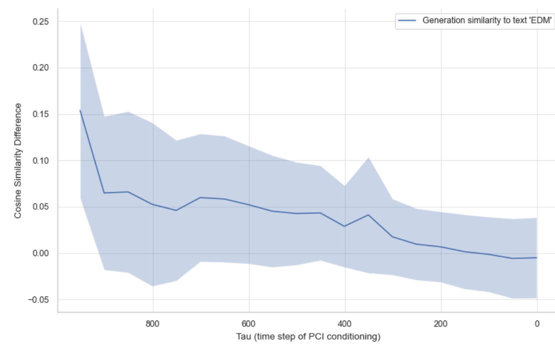


(b) Band context

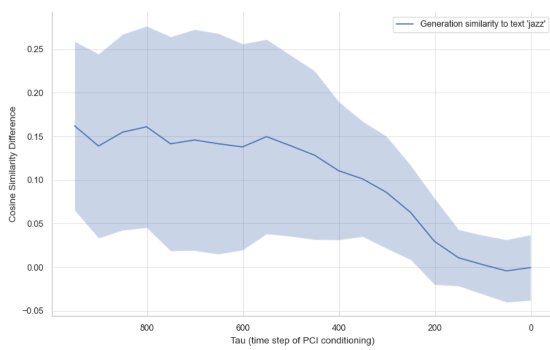
Figure 4.8: Success Rate curves across diffusion steps, averaged across instrument concepts



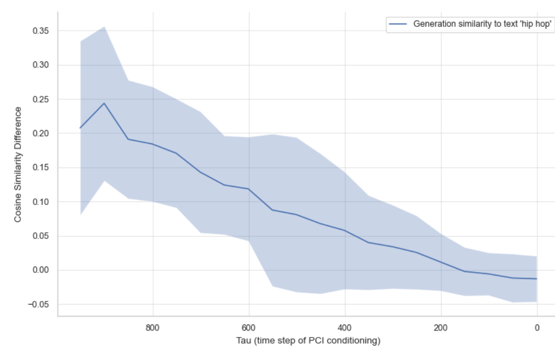
(a) Concept 'classical'



(b) Concept 'EDM'



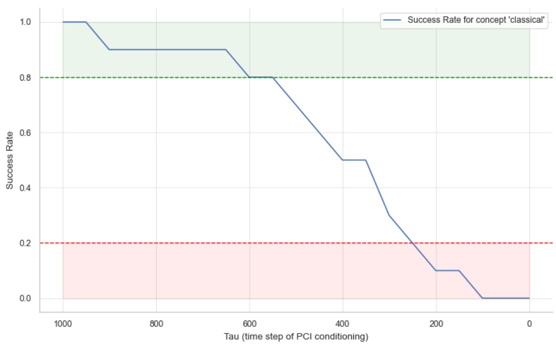
(c) Concept 'jazz'



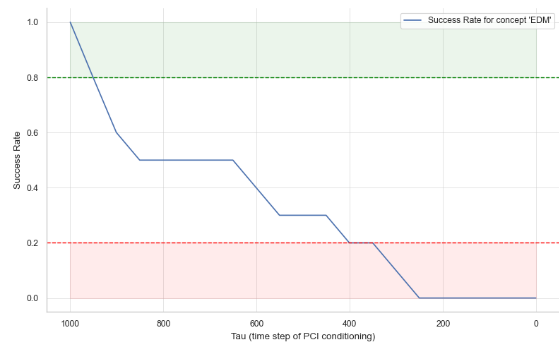
(d) Concept 'hip hop'

Figure 4.9: Mean activation curves (eq.3.14 and eq.3.15) across diffusion steps for genre concepts.

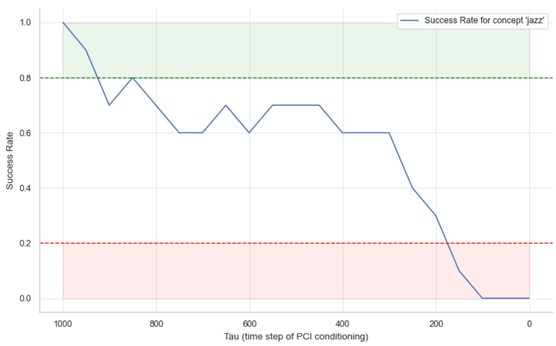
(Figure 4.11 right), the high-success window is relatively narrow, centered around $\tau \approx 900$. The failure region emerges once τ decreases below approximately 250. This behavior can be attributed to the fact that, unlike instrumental concept insertion, genre plays a fundamental role in determining the overall structure of a musical piece. Since this structure is largely established during the early diffusion steps, later interventions become less effective in modifying it.



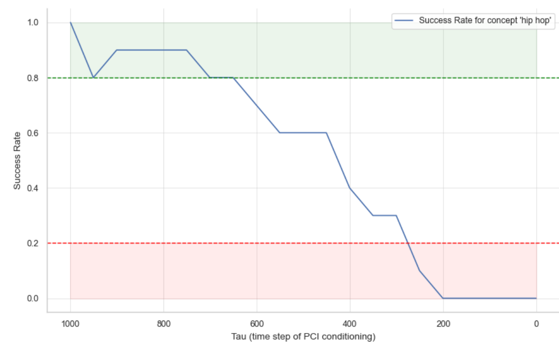
(a) Concept 'classical'



(b) Concept 'EDM'



(c) Concept 'jazz'



(d) Concept 'hip hop'

Figure 4.10: Success rate curves across diffusion steps for genre concepts.

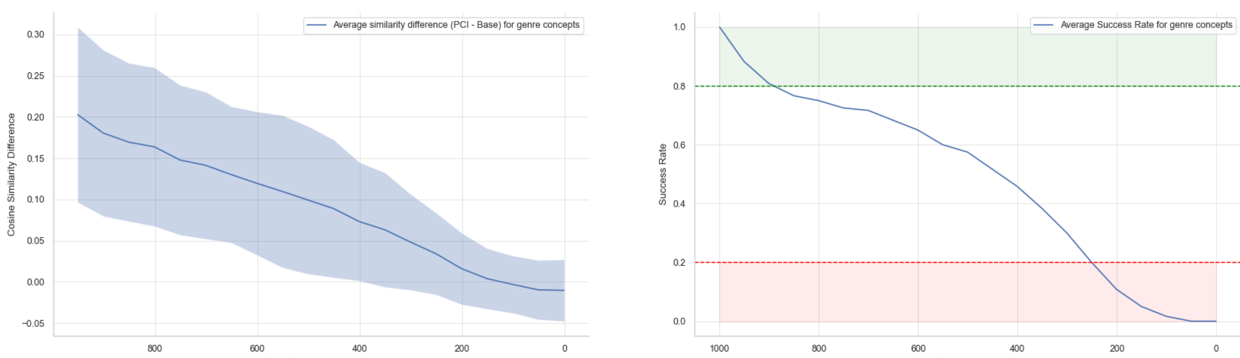
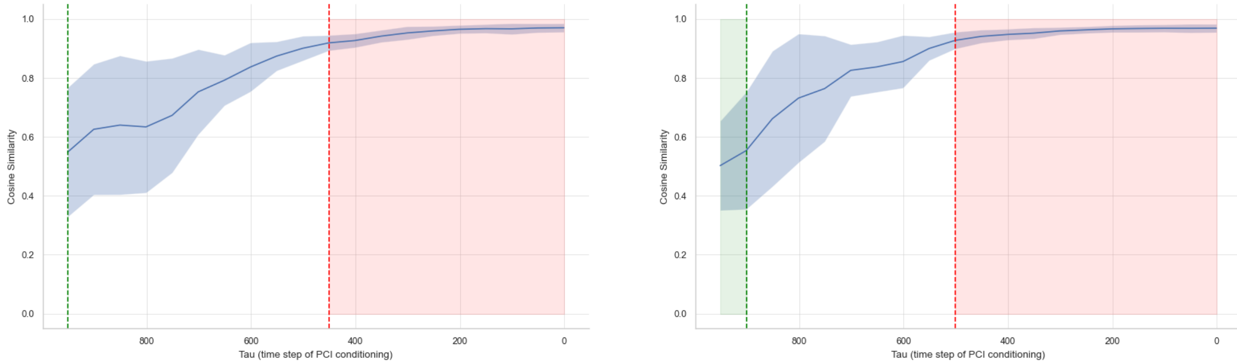


Figure 4.11: Mean activation curve (left) and Success Rate curve (right) across diffusion steps averaged across genre concepts.

Temporal Editability Analysis

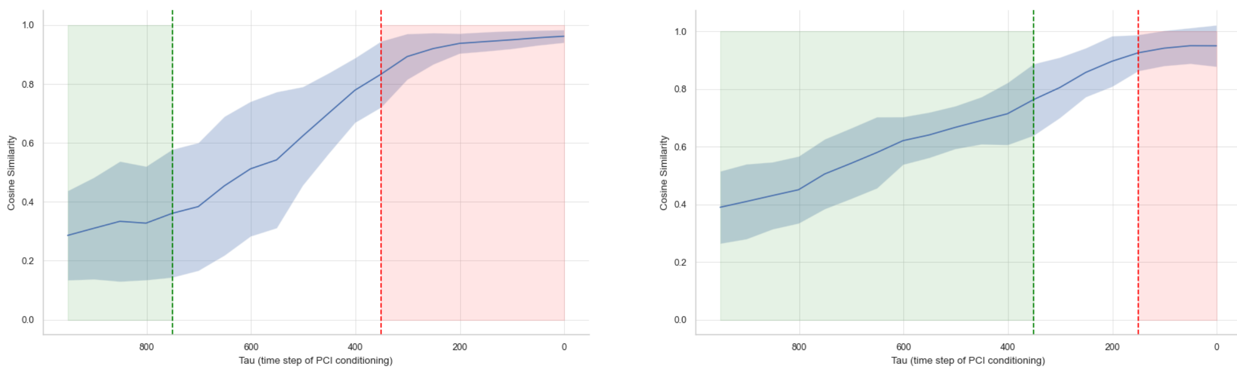
Instrumentation. As discussed previously, instrument concept insertion in the orchestral context is limited to a short period of time ($\tau > 950$). Within this high-success window (highlighted in green) of the corresponding mean base-similarity curve (Figure 4.12), the similarity values range between 0.5 and 0.6. This indicates that the base generation undergoes substantial modification in order to successfully inject the *piano* and *drum* concepts into the orchestral context.

Considering the instrument concepts injected in the *band* context (Figure 4.13), a much wider success windows can be observed. However, a concept-wise analysis reveals that for the *violin*, the

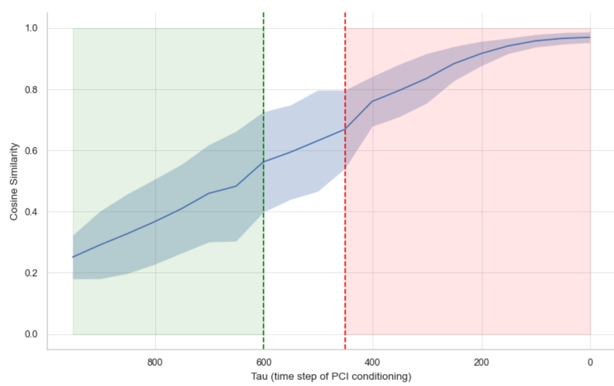


(a) Caption 'drum' (b) Caption 'piano'

Figure 4.12: Mean base similarity curves (eq. 3.20 and eq. 3.21) across diffusion steps for instrument concepts in an *orchestral* context.



(a) Concept 'violin' (b) Concept 'trumpet'



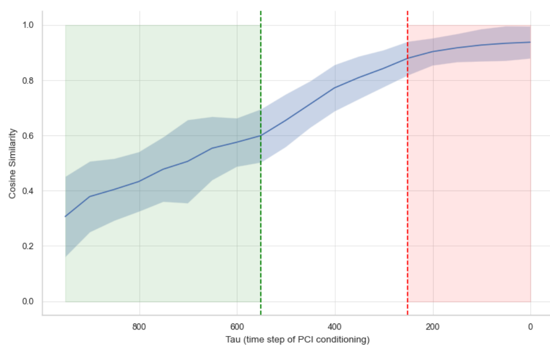
(c) Concept 'flute'

Figure 4.13: Mean base similarity curves (eq. 3.20 and eq. 3.21) across diffusion steps for instrument concepts in a *band* context.

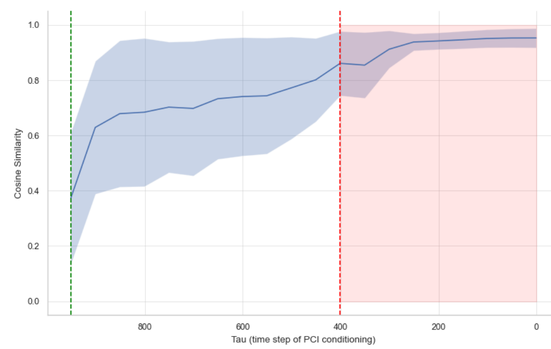
base similarity drops considerably within the success region (falling below 0.4). Manual inspection of the corresponding audio samples indicates that, although the *violin* concept is clearly perceptible, the underlying *band* context is only weakly preserved within this window. A plausible explanation is that the training data of the generative model contains relatively few examples where a *violin* is performed in a live band setting, making such combinations harder to synthesize faithfully.

Similar to the *piano* and *drum* cases in the *orchestral* context, the base similarity after injecting the *flute* drops below 0.6 within the high-success region. This indicates that the original generation is modified considerably in order to incorporate the concept.

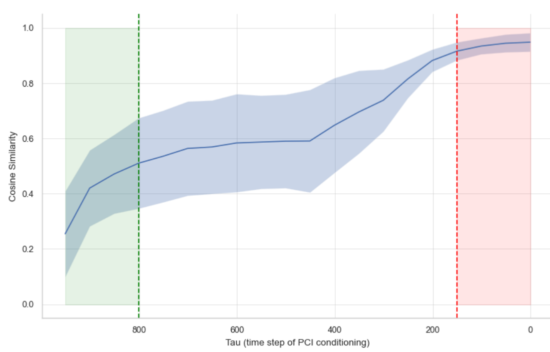
In contrast, for the *trumpet*, the base similarity remains relatively high (ranging between 0.6 and 0.8) within the success window until $\tau \approx 600$, providing a favorable temporal window for editable intervention.



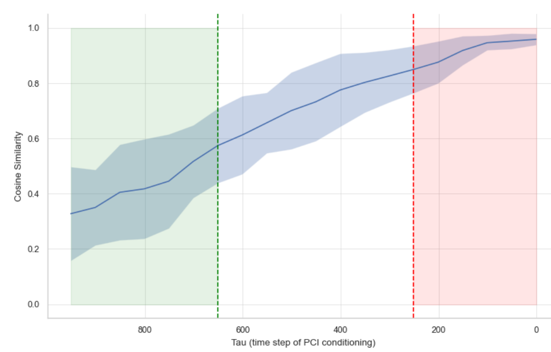
(a) Concept ’classical’



(b) Concept ’EDM’



(c) Concept ’jazz’



(d) Concept ’hip hop’

Figure 4.14: Mean base similarity curves (eq. 3.20 and eq. 3.21) across diffusion steps for genre concepts.

Genre. For all *genre* concepts (Figure 4.14), the base similarity remains below 0.6 within the high-success window, indicating that the original generation is substantially modified to incorporate the genre. This behavior is expected, as altering the genre of a musical piece is fundamentally different from editing a single instrumental stem. Genre determines the overall instrumentation, rhythm, tempo, and harmonic structure of the music. Consequently, such deviations from the base prompt generation are both reasonable and necessary.

Chapter 5

Conclusions & Future Works

This work sheds light on the under-explored area of temporal concept dynamics in music generation models. Two approaches were employed: intermediate latent analysis for different concepts, and Prompt Conditioned Intervention (PCI), recently used for the same purpose in image generation models.

The first method did not provide sufficient insight into nuanced temporal behaviors. In contrast, PCI revealed important dynamics. Instrumentation concepts could be successfully injected even at later diffusion time steps, whereas genre concepts were only successfully incorporated within an early narrow diffusion window. This is expected, as genre largely determines the overall structure of a musical piece. Furthermore, PCI dynamics were observed to be highly sensitive to the context of the base prompt, as evident from the differences in instrumentation concept insertion in the two different musical contexts.

Considering editability, only the trumpet showed a sufficient diffusion window where the concept could be successfully incorporated while keeping minimal changes to the initial generation. These conclusions may differ with different base prompt contexts; therefore, a thorough analysis with varied contexts for each fine-grained concept is needed in future work.

Due to computational constraints, the success of concept insertion in this study was determined via thresholding on CLAP similarities. This approach can be sensitive to the chosen thresholds, even though they were selected empirically for each concept category. A more robust evaluation could be achieved using existing audio Q&A models [14, 15], although these require significantly more computational resources.

.1 Appendix

Prompt Pairs: Tempo

- **tempo_1**
  A: A *slow*, atmospheric ambient piano track with soft reverb and long sustained notes
  B: A *fast*, atmospheric ambient piano track with soft reverb and long sustained notes

- **tempo_2**
  A: A *slow* electronic music piece with a smooth texture and evolving background pads
  B: A *fast* electronic music piece with a smooth texture and evolving background pads

- **tempo_3**
  A: A *slow* cinematic soundtrack featuring gentle melodies and a calm mood
  B: A *fast* cinematic soundtrack featuring gentle melodies and a calm mood

- **tempo_4**
  A: A *slow* lo-fi instrumental track with warm tones and a relaxed feeling
  B: A *fast* lo-fi instrumental track with warm tones and a relaxed feeling

- **tempo_5**
  A: A *slow* ambient soundscape designed for meditation and focus
  B: A *fast* ambient soundscape designed for meditation and focus

Prompt Pairs: Instrumentation

- **instrument_1**
  A: An ambient instrumental track led by a soft *piano* melody with minimal accompaniment
  B: An ambient instrumental track led by a soft *acoustic guitar* melody with minimal accompaniment

- **instrument_2**
  A: A cinematic music piece featuring sustained *string* instruments and a spacious atmosphere
  B: A cinematic music piece featuring sustained *brass* instruments and a spacious atmosphere

- **instrument_3**
  A: An electronic ambient track dominated by *warm analog synthesizer* sounds
  B: An electronic ambient track dominated by *bright digital synthesizer* sounds

- **instrument_4**
  A: A calm instrumental piece centered around a solo *piano* with subtle background textures
  B: A calm instrumental piece centered around a solo *flute* with subtle background textures

- **instrument_5**
  A: A minimalist ambient track built around *deep, resonant bass tones*
  B: A minimalist ambient track built around *light, airy high-frequency tones*

Prompt Pairs: Rhythm

- **rhythm_1**
  A: An electronic track with a *steady and regular* rhythmic pattern throughout
  B: An electronic track with a *syncopated and irregular* rhythmic pattern throughout
- **rhythm_2**
  A: A percussion-based music piece with a *consistent and predictable* beat
  B: A percussion-based music piece with a *complex and unpredictable* beat
- **rhythm_3**
  A: An ambient electronic track with a *smooth and continuous* rhythm
  B: An ambient electronic track with a *broken and fragmented* rhythm
- **rhythm_4**
  A: A rhythmic instrumental track with *evenly spaced* drum hits
  B: A rhythmic instrumental track with *unevenly spaced* drum hits
- **rhythm_5**
  A: A calm electronic music piece with a subtle and *repetitive* rhythmic pulse
  B: A calm electronic music piece with a subtle but *constantly shifting* rhythmic pulse

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